3.3.3 Higher Order Symplectic Methods: The Suzuki-Yoshida Method

Yoshida [398] gives an elegant method for creating a symplectic scheme of arbitrarily high order, using Trotter's results for compositions of linear operators [370]. This work is related to methods suggested by Suzuki [354] in the context of Trotter factorization of quantum operators. Consider a scheme with order $2s$ (for $s \geq 1$) and where the evolution of the system under the method is given by $\exp(h\,\hat{\tilde{\mathcal{L}}}_h)$, where

$$\hat{\mathcal{L}}_h \stackrel{\text{def}}{=} \mathcal{L}_{\tilde{H}_h} = \mathcal{L}_H + h^{2s}\tilde{\mathcal{L}}_{[2s]} + h^{2s+2}\tilde{\mathcal{L}}_{[2s+2]} + h^{2s+4}\tilde{\mathcal{L}}_{[2s+4]} + \mathcal{O}(h^{2s+6})$$

is the method's characteristic operator (with corresponding shadow Hamiltonian $\tilde{H}_h$). The linear operators $\tilde{\mathcal{L}}_{[2i]}$ are compositions of Lie derivatives, and can be computed using the symmetric BCH formula. We now consider iterating this scheme three times, first using a step of $\tau_0 h$, followed by a step of $\tau_1 h$, and finally another step with stepsize $\tau_0 h$ (where $\tau_0 h + \tau_1 h + \tau_0 h = h$). The overall effect on the system is given by a product of exponentials, as

$$\exp\left(\tau_0 h\,\hat{\mathcal{L}}_{\tau_0 h}\right)\exp\left(\tau_1 h\,\hat{\mathcal{L}}_{\tau_1 h}\right)\exp\left(\tau_0 h\,\hat{\mathcal{L}}_{\tau_0 h}\right) = \exp\left(h\,Z_h\right),$$

where, using repeated application of the BCH formula, we find

$$\begin{aligned} Z_h &= 2\tau_0\,\hat{\mathcal{L}}_{\tau_0 h} + \tau_1\,\hat{\mathcal{L}}_{\tau_1 h} + \mathcal{O}(h^{2s+2}), \\ &= (2\tau_0 + \tau_1)\,\mathcal{L}_H + (2\tau_0^{2s+1} + \tau_1^{2s+1})h^{2s}\tilde{\mathcal{L}}_{[2s]} + \mathcal{O}(h^{2s+2}). \end{aligned}$$

We have free reign over constants τ_0 and τ_1 as long as $2\tau_0 + \tau_1 = 1$. Hence we have an opportunity to annihilate the perturbation operator at order h^{2s} by choosing $2\tau_0^{2s+1} + \tau_1^{2s+1} = 0$ as well. Solving simultaneously, there exists a unique real solution

$$\tau_0 = \frac{1}{2-\kappa}, \qquad \tau_1 = -\frac{\kappa}{2-\kappa}, \qquad \kappa^{2s+1} = 2,$$

giving us a scheme of order $2s+2$. We can then proceed recursively, as this new order $2s+2$ scheme can be composed similarly to wipe out successive higher order terms.

Though the fourth-order version of the scheme was given first by Forest and Ruth [139] (and discovered independently by Yoshida [398] and Candy and Rozmus [66]), we shall simply refer to these higher-order schemes as Yoshida methods, owing to the elegant derivation of schemes of arbitrary order.

Example 3.1 Consider using velocity Verlet as the base second-order method to build a Yoshida fourth-order method from. As the scheme is second-order, we have

$s = 1$, and hence

$$\tau_0 = \frac{1}{2 - \sqrt[3]{2}}, \qquad \tau_1 = -\frac{\sqrt[3]{2}}{2 - \sqrt[3]{2}}.$$

The overall scheme is then three iterations of velocity Verlet, using stepsizes $\tau_0 h$, $\tau_1 h$ and $\tau_0 h$ respectively. We write this with subindices α, β to indicate the intermediate stages.

Yoshida fourth-order scheme (velocity Verlet):

$$\begin{aligned}
\boldsymbol{P}_\alpha &\coloneqq \boldsymbol{p} - (\tau_0 h/2)\, \nabla U(\boldsymbol{q}), \\
\boldsymbol{Q}_\alpha &\coloneqq \boldsymbol{q} + (\tau_0 h)\, \boldsymbol{M}^{-1} \boldsymbol{P}_\alpha, \\
\boldsymbol{P}_\alpha &\coloneqq \boldsymbol{P}_\alpha - (\tau_0 h/2)\, \nabla U(\boldsymbol{Q}_\alpha), \\
\boldsymbol{P}_\beta &\coloneqq \boldsymbol{P}_\alpha - (\tau_1 h/2)\, \nabla U(\boldsymbol{Q}_\alpha), \\
\boldsymbol{Q}_\beta &\coloneqq \boldsymbol{Q}_\alpha + (\tau_1 h)\, \boldsymbol{M}^{-1} \boldsymbol{P}_\beta, \\
\boldsymbol{P}_\beta &\coloneqq \boldsymbol{P}_\beta - (\tau_1 h/2)\, \nabla U(\boldsymbol{Q}_\beta), \\
\boldsymbol{P} &\coloneqq \boldsymbol{P}_\beta - (\tau_0 h/2)\, \nabla U(\boldsymbol{Q}_\beta), \\
\boldsymbol{Q} &\coloneqq \boldsymbol{Q}_\beta + (\tau_0 h)\, \boldsymbol{M}^{-1} \boldsymbol{P}, \\
\boldsymbol{P} &\coloneqq \boldsymbol{P} - (\tau_0 h/2)\, \nabla U(\boldsymbol{Q}).
\end{aligned} \tag{3.8}$$

The equations could easily be written in a simplified form, combining several of the steps. This scheme requires three new evaluations of the force ∇U per iteration, making it significantly more expensive than the vanilla second-order Verlet method. We may use this fourth-order scheme (now $s = 2$) to then find a scheme of order $2s + 2 = 6$.

We compare the symplectic methods developed so far in Fig. 3.2, for $t \in [0, 100]$ using planar phase space and the Lennard-Jones oscillator (the one degree-of-freedom problem with Hamiltonian $H(q, p) = p^2/2 + \varphi_{\mathrm{LJ}}(q)$). As anticipated, there is no visible drift in the Hamiltonian as the simulation time increases. In each case, we observe an oscillation in the fluctuation of the total energy. Additionally we compute trajectories propagated at different stepsizes, and plot the maximum deviation

$$\max |H(q_k, p_k) - H(q_0, p_0)|$$

over the entire trajectory. We can see that for an order s method, the maximum deviation in the Hamiltonian is of order s, as is expected from the analysis. Yoshida fourth-order schemes are tested using both the position and velocity Verlet schemes